MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Core, Think Harder and Challenge questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 11-12 (finishing P5)

Math Inquirers P5 Ages 11-12. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three levels. Every child starts with Core, then moves up when they are ready. Challenge questions are the ones to talk about together.
- CORE builds fluency with the main idea of the lesson. Use it for warm-ups and for children who need confidence.
- THINK HARDER goes a step further: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- CHALLENGE questions are non-routine. They ask children to find every way, work backwards, test a case, justify an answer or notice what is missing. This is the style of thinking used in gifted-education-type problems. It is not an official GEP standard, and this program is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Million Puzzles (Numbers to 10 Million)	6 + 8 + 9 questions
Lesson 2	Fraction Feats (Fractions: Four Operations)	6 + 8 + 9 questions
Lesson 3	Before and After (Fraction Word Problems)	6 + 8 + 9 questions
Lesson 4	Triangle Areas (Area of Triangle)	6 + 8 + 9 questions
Lesson 5	Volume Vault (Volume)	6 + 8 + 9 questions
Lesson 6	Rate Racers (Rate)	6 + 8 + 9 questions
Lesson 7	Percentage Puzzles (Percentage)	6 + 8 + 9 questions
Lesson 8	Angle Chasers (Angles, Triangles and Quadrilaterals)	6 + 8 + 9 questions
Lesson 9	Pattern Generalisers (Patterns and Number Sense)	6 + 8 + 9 questions
Lesson 10	Challenge Day (Mixed Heuristics)	6 + 8 + 9 questions

Lesson 1: Million Puzzles

Numbers to 10 Million • Week 1

CORE

C1. Write 4 030 205 in words.

C2. What is the value of the digit 6 in 3 621 480?

C3. Round 5 482 716 to the nearest thousand.

C4. Work out 2 750 000 ÷ 1000.

C5. Work out 48 − 6 × (3 + 4).

C6. Estimate 6912 × 48 by rounding 6912 to the nearest thousand and 48 to the nearest ten.

THINK HARDER

T1. How many 7-digit numbers have digits that add up to 2? List them all.

T2. Put ONE pair of brackets into 2 + 3 × 4 + 5 to make 25. Then put one pair in a different place to make 29.

T3. A whole number rounded to the nearest thousand is 640 000. What is the greatest it could be? The smallest?

T4. Use each of the digits 1, 2, 3, 4, 5, 6 and 7 once. What is the difference between the greatest and the smallest 7-digit number you can make?

T5. Which is the best estimate of 4108 × 29: 12 000, 120 000 or 1 200 000? Explain.

T6. Siti saves $250 every week. How many weeks will it take her to save $1 000 000? About how many years is that?

T7. Use four 4s with +, −, ×, ÷ and brackets to make 3, then 5, then 7.

T8. What is the smallest 7-digit number whose digits are all different? The greatest?

CHALLENGE (non-routine: justify, find every way)

X1. How many 7-digit numbers are there altogether?

X2. Write 1 000 000 as the product of two whole numbers that have no zero digits at all.

X3. Ben multiplies a 4-digit number by a 3-digit number. How many digits can his answer have?

X4. A palindrome reads the same forwards and backwards, like 3 456 543. How many 7-digit palindromes are there between 3 000 000 and 4 000 000?

X5. Use 1, 2, 3, 4, 5 in this order, with +, −, ×, ÷ and brackets, to make 100.

X6. A heart beats about 70 times a minute. About how many days does it take to beat 1 000 000 times?

X7. Put brackets into 1 + 2 × 3 + 4 × 5 + 6 to make the answer as big as possible. You may use more than one pair. What is the biggest answer?

X8. I am a 7-digit odd number. My digits are all different and add up to 42. Rounded to the nearest hundred thousand, I am 7 000 000. I am the smallest number that fits. Who am I?

X9. Ali writes all the whole numbers from 1 to 1000. How many digits does he write? (Go further: how many if he writes all the numbers from 1 to 1 000 000?)

Lesson 2: Fraction Feats

Fractions: Four Operations • Week 1

CORE

C1. Write 7 ÷ 9 as a fraction.

C2. Work out 2/3 + 3/4.

C3. Work out 5/6 − 1/4.

C4. Find 3/5 of 15.

C5. Work out 2/3 × 9/10.

C6. Work out 3/5 × 1/2 + 1/10.

THINK HARDER

T1. Write 2/3 as the sum of two different unit fractions.

T2. Ben ate 1/3 of a pizza. Ali ate 1/4 of what was left. What fraction of the pizza is left?

T3. 1/2 − 1/3 = 1/6 and 1/3 − 1/4 = 1/12. Use the pattern to work out 1/4 − 1/5 and 1/9 − 1/10.

T4. 4 litres of juice are poured equally into 6 bottles. How much juice is in each bottle?

T5. Which is bigger, 5/6 or 6/7? Explain without finding common denominators.

T6. A rope 3/4 m long is cut into 3 equal pieces. How long is each piece?

T7. Find a fraction between 2/5 and 1/2 with 20 as the bottom number.

T8. Which is bigger, 7/8 of 24 or 5/6 of 30?

CHALLENGE (non-routine: justify, find every way)

X1. Write 1 as the sum of three different unit fractions. Find all the ways.

X2. Work out 1/2 × 2/3 × 3/4 × 4/5 × … × 99/100.

X3. Mrs Tan baked some cookies. She gave 1/2 of them to her neighbours, 1/3 of the remainder to Ben and 1/4 of what was then left to Mei. She had 30 cookies left. How many did she bake?

X4. Which is more: 2/3 of 3/4 of a cake, or 3/4 of 2/3 of a cake?

X5. Use the digit cards 1, 3, 5 and 7 once each to make two fractions. Take the smaller from the bigger. What is the smallest difference you can make (more than 0)?

X6. Mei spent 1/4 of her money on Monday and 1/3 of the remainder on Tuesday. She had $50 left. How much did she spend each day?

X7. Work out 1/6 + 1/12 + 1/20 + 1/30 + 1/42 without finding a common denominator for all five. (Hint: 1/6 = 1/2 − 1/3.)

X8. 5 pizzas are shared equally among 12 children. Each child must get exactly 2 pieces, and each piece must be a unit fraction of a pizza (like 1/3 or 1/4). How can the pizzas be cut?

X9. A tank is full of water. 1/3 of the water is let out. Then 1/3 of the water left is let out, and then 1/3 of the water left again. Now 16 litres are left. How much water was in the full tank?

Lesson 3: Before and After

Fraction Word Problems • Week 2

CORE

C1. A baker made some buns. She sold 2/5 of them in the morning and 1/3 of the remaining buns in the afternoon. She had 120 buns left. How many buns did she make?

C2. Ben is 10 years old and his father is 38. In how many years will Ben's age be 1/2 of his father's age?

C3. Siti had $60. She spent 1/4 of it on a book and 1/3 of the remainder on a T-shirt. How much money did she have left?

C4. Ali had 26 stamps and Ravi had 34. How many stamps must Ravi give Ali so that they have the same number?

C5. There are 40 pupils in a hall. 3/8 of them are boys. How many more girls than boys are there?

C6. Mei and Siti had the same amount of money. Mei spent $18 and Siti spent $42. Then Siti had 1/3 as much money as Mei. How much did each of them have at first?

THINK HARDER

T1. Ravi spent 1/4 of his money on Monday and 2/5 of the remainder on Tuesday. He spent $66 altogether. How much money did he have at first?

T2. A box had red pens and blue pens. 2/3 of the pens were blue. After 12 blue pens were taken out, 1/2 of the pens were blue. How many pens were in the box at first?

T3. Mrs Tan is 41 years old and her son is 9. How many years ago was her son's age 1/5 of her age?

T4. Ben and Siti had $140 altogether. Ben spent 1/2 of his money and Siti spent $20. Then they had the same amount left. How much did Ben have at first?

T5. Ali had 40 more stickers than Mei. Each of them was given 10 more stickers. Then Mei had 3/5 as many stickers as Ali. How many stickers did Mei have at first?

T6. A tank was 3/4 full of water. After 18 litres were used, it was 1/2 full. How many litres does the tank hold when it is full?

T7. Siti gave 1/3 of her stickers to Mei. Then they had the same number of stickers. Mei had 15 stickers at first. How many did Siti have at first?

T8. Ravi had $45 and Ben had $15. Their aunt gave each of them the same amount of money. Then Ben had 1/2 as much as Ravi. How much did the aunt give each boy?

CHALLENGE (non-routine: justify, find every way)

X1. Ali, Ben and Mei shared $180. Ali took 1/3 of the money, Ben took 1/4 of what was left and Mei took the rest. How much more did Mei get than Ali?

X2. Mei had some money. She spent $12, then spent 1/2 of what was left, then spent $8. She had $10 left. How much did she have at first?

X3. A father and his son are 50 years old altogether. In 5 years, the son's age will be 1/3 of the father's age. How old is the son now?

X4. Ali had 3/5 as many cards as Ben. After Ben gave Ali 20 cards, they had the same number. How many cards did they have altogether?

X5. In a basket, 1/3 of the fruits were apples. 10 apples were eaten and 10 pears were put in. Now 1/6 of the fruits are apples. How many fruits are in the basket?

X6. Siti spent 2/5 of her money on a dress and 1/3 of the remainder on shoes. The dress cost $30 more than the shoes. How much money did Siti have at first?

X7. Mei had $200 and Siti had $80. Mei spent 3 times as much money as Siti. Then they had the same amount left. How much money did Siti spend?

X8. Ravi had some stickers. He gave 1/2 of them and 4 more to Ali. Then he gave 1/2 of the stickers he had left and 4 more to Ben. He had 4 stickers left. How many stickers did he have at first?

X9. Ali and Ben had 140 stamps altogether. Ali gave away 1/3 of his stamps and Ben gave away 1/2 of his. Then they had the same number of stamps left. How many stamps did Ben give away?

Lesson 4: Triangle Areas

Area of Triangle • Week 2

CORE

C1. A triangle has a base of 12 cm and a height of 5 cm. What is its area?

C2. A right-angled triangle has sides of 8 cm, 15 cm and 17 cm. What is its area?

C3. A triangle has an area of 36 cm² and a height of 8 cm. How long is its base?

C4. A rectangle 9 cm by 6 cm is cut along a diagonal into two triangles. What is the area of each triangle?

C5. A square has sides of 8 cm. A triangle inside it has one side of the square as its base and its top corner on the opposite side. What is the area of the triangle?

C6. A triangle has a base of 14 cm and a height of 9 cm. A second triangle has the same base and the same height. What is the total area of the two triangles?

THINK HARDER

T1. A triangle has an area of 45 cm². Its base is 10 cm. What is its height?

T2. A right-angled triangle has sides of 12 cm, 16 cm and 20 cm. What is its area?

T3. A square has sides of 10 cm. A triangle is drawn with one side of the square as its base and its top corner on the opposite side. What is the area of the triangle?

T4. A triangle and a rectangle have the same base, 12 cm, and the same height, 7 cm. What is the difference between their areas?

T5. Triangle P has a base of 8 cm and a height of 5 cm. Triangle Q has a base of 4 cm and the same area as P. What is the height of Q?

T6. A rectangle is 12 cm by 8 cm. M is the middle of the top side. M is joined to the two bottom corners. What is the area of the triangle? What is the area of the rest of the rectangle?

T7. A figure is a square of side 6 cm with a triangle on top. The base of the triangle is the top side of the square and its height is 4 cm. What is the area of the figure?

T8. What is the area of a triangle with a base of 7.5 cm and a height of 4 cm?

CHALLENGE (non-routine: justify, find every way)

X1. Triangle ABC has an area of 60 cm². D is the middle point of BC. What is the area of triangle ABD?

X2. A rectangle is 12 cm by 8 cm. P is the middle of the top side and Q is the middle of the right side. What is the area of the triangle joining P, Q and the bottom-left corner?

X3. A triangle and a square have the same area. The square has sides of 6 cm. The triangle has a base of 9 cm. What is the height of the triangle?

X4. Square ABCD has sides of 8 cm (A top-left, B bottom-left, C bottom-right, D top-right). BC is extended to E so that CE = 4 cm. AE cuts CD at F. What is the difference between the area of triangle ADF and the area of triangle FCE?

X5. List every pair of whole numbers (base and height, in cm) for a triangle with an area of 6 cm².

X6. Ali draws a triangle with a base of 10 cm and a height of 6 cm. Mei draws a triangle with a base of 5 cm and twice the area of Ali's triangle. What is the height of Mei's triangle?

X7. A right-angled triangle has sides of 15 cm, 20 cm and 25 cm. What is its height when the 25 cm side is the base?

X8. Two squares stand side by side on a line, touching: a big square of side 10 cm and a small square of side 6 cm. Join the bottom-left corner of the big square, the top-right corner of the big square and the top-right corner of the small square to make a triangle. What is its area? Try again with a small square of side 4 cm. What do you notice?

X9. Triangle ABC has an area of 54 cm². D is a point on BC with BD = 4 cm and DC = 8 cm. What is the area of triangle ABD?

Lesson 5: Volume Vault

Volume • Week 3

CORE

C1. A cuboid is 9 cm long, 4 cm wide and 5 cm high. What is its volume?

C2. A cube has edges of 6 cm. How many 1-cm cubes are needed to build it?

C3. A tank 40 cm long, 20 cm wide and 30 cm high is half full of water. How many litres of water are in it?

C4. A cuboid has a volume of 240 cm³. Its base is 8 cm by 6 cm. How high is it?

C5. A tank with a base 25 cm by 20 cm has water 6 cm deep. 2 ℓ more water is poured in. How deep is the water now?

C6. A stone sinks to the bottom of a tank with a base 20 cm by 15 cm. The water rises 2 cm. What is the volume of the stone?

THINK HARDER

T1. How many cubes with 2-cm edges can be packed in neat layers into a box 9 cm long, 6 cm wide and 5 cm high?

T2. A tank is 1/3 full of water. After 12 ℓ more is poured in, it is 3/5 full. How many litres does the tank hold when full?

T3. A cuboid has a square base and a height of 5 cm. Its volume is 245 cm³. How long is each side of the base?

T4. A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have no paint at all?

T5. A tank is 60 cm long, 40 cm wide and 50 cm high. A bucket holds 8 ℓ. What is the fewest number of buckets of water needed to fill the tank at least 3/4 full?

T6. A tank has a base 40 cm by 30 cm. It holds 36 ℓ of water. How deep is the water?

T7. A cube-shaped container with 20-cm edges is full of water. All of it is poured into an empty tank with a base 50 cm by 32 cm. How deep is the water in the tank?

T8. A cube with 10-cm edges is cut into small cubes with 2-cm edges. How many small cubes are there?

CHALLENGE (non-routine: justify, find every way)

X1. A big cube is painted on the outside and cut into small cubes of the same size. 96 small cubes have exactly one painted face. How many small cubes are there altogether?

X2. How many different cuboids can be built with exactly 64 one-centimetre cubes, using all of them each time?

X3. A tank has a base 40 cm by 30 cm. A cube with edges of 10 cm sits on the bottom, and the water is 6 cm deep around it. How much water is in the tank?

X4. Water in a tank is 15 cm deep. When 4 identical bricks are put in, the water rises to 18 cm. 6 more of the same bricks are put in. How deep is the water now? (The bricks stay under water.)

X5. A 5 by 5 by 5 cube is built from 1-cm cubes. All the cubes that cannot be seen from outside are taken out, so it is hollow. How many cubes are left?

X6. Bricks measure 2 cm by 3 cm by 4 cm. They are all placed the same way up to build a solid cube. What is the fewest number of bricks needed?

X7. Three faces of a cuboid have areas 12 cm², 15 cm² and 20 cm². What is the volume of the cuboid?

X8. A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have exactly 2 painted faces? How many have exactly 1?

X9. A tank has a base 30 cm by 20 cm and water 10 cm deep. A block 10 cm by 10 cm by 15 cm is stood upright on the bottom. How deep is the water now?

Lesson 6: Rate Racers

Rate • Week 3

CORE

C1. 5 kg of flour cost $8.50. How much do 3 kg cost?

C2. A printer prints 24 pages a minute. How long does it take to print 360 pages?

C3. Which is the better buy: 4 rolls for $3.40 or 6 rolls for $4.98?

C4. A tap fills a 120 ℓ tank at 8 ℓ a minute. How long does it take?

C5. Two taps fill 10 ℓ and 14 ℓ a minute. How long do they take together to fill a 192 ℓ tank?

C6. Nuts cost $1.50 for every 100 g. How much do 650 g cost?

THINK HARDER

T1. A tap fills a tank in 12 minutes at 15 ℓ a minute. When a second tap is also turned on, the tank fills in 9 minutes. How many litres a minute does the second tap fill?

T2. Eggs are sold in boxes of 10 for $3.30 and boxes of 6 for $2.10. Siti needs at least 28 eggs. What is the least she can pay?

T3. A car park charges $1.50 for the first hour and $1 for every half hour or part of a half hour after that. How much does it cost to park from 10.15 a.m. to 2.50 p.m.?

T4. Phone plan A costs $15 a month plus 10¢ a minute. Plan B costs 25¢ a minute with no monthly fee. For how many minutes in a month do the two plans cost the same?

T5. 3 machines make 360 masks in 4 minutes. At the same rate, how many masks do 5 machines make in 6 minutes?

T6. Ali earns $9 an hour on weekdays and $12 an hour on Saturdays. He works 4 hours each weekday from Monday to Friday and 5 hours on Saturday. How much does he earn in the week?

T7. A full tank holds 300 ℓ. A drain lets out 15 ℓ a minute, but a tap is left on that adds 5 ℓ a minute. How long does the tank take to empty?

T8. A car uses 8 ℓ of petrol for every 100 km. How much petrol does it use for 350 km?

CHALLENGE (non-routine: justify, find every way)

X1. Printer P prints a job in 6 minutes. Printer Q prints the same job in 3 minutes. How long do they take working together?

X2. Siti buys pens at 3 for $2 and sells them at 2 for $2. How many pens must she buy and sell to make a profit of $10?

X3. A machine starts at 9.52 a.m. and makes 50 cups a minute. At 10.00 a.m. a second machine starts, making 30 cups a minute. At what time have they made 1600 cups altogether?

X4. Ben needs at least 20 cans. Shop A sells 6 cans for $4.20, shop B sells 4 cans for $2.92 and shop C sells a deal of 10 cans for $7.80 with 2 cans free. He can buy from more than one shop, but only whole packs. What is the least he can pay?

X5. A tap fills 9 ℓ a minute. How long does it take to fill an empty tank 60 cm long, 50 cm wide and 45 cm high?

X6. Ravi reads 24 pages an hour and Siti reads 18 pages an hour. They start the same 360-page book at the same time. When Ravi finishes, how many pages does Siti still have to read?

X7. 3 cats eat 3 fish in 3 minutes. At this rate, how many cats are needed to eat 100 fish in 100 minutes?

X8. Candle A is 20 cm tall and burns down 2 cm every hour. Candle B is 14 cm tall and burns down 1 cm every hour. They are lit at the same time. After how many hours are they the same height? How tall are they then?

X9. A tap fills an empty bath in 12 minutes. With the plug out, a full bath empties in 20 minutes. Ben turns on the tap but forgets the plug. How long does the bath take to fill?

Lesson 7: Percentage Puzzles

Percentage • Week 4

CORE

C1. Find 10% of $360 and 15% of $360.

C2. A book costs $45. It is sold at a 20% discount. What is the sale price?

C3. A meal costs $30 before GST. GST is 9%. How much is the meal with GST?

C4. 24 of the 40 pupils in a class walk to school. What percentage walk to school?

C5. Ali had $60. He spent 25% of it. How much did he have left?

C6. 21 of the 60 seats in a hall are empty. What percentage of the seats are empty?

THINK HARDER

T1. At a hawker centre, a plate of chicken rice went up from $4 to $5. A month later, it went back down from $5 to $4. (a) Find the percentage increase from $4 to $5. (b) Find the percentage decrease from $5 to $4. (c) Both changes are $1. Why are the two percentages different? Hint: percentage change = the change out of the amount you START with.

T2. A box of cereal costs $15 in two shops. Shop X: "Buy 3, get 1 free!" Shop Y: "30% off every box!" Mei buys 4 boxes. (a) How much does she pay at each shop? Which is the better buy, and by how much? (b) Shop X's deal is really how many percent off?

T3. Siti says, "15% of $80 is $8 + $4 = $12." Explain her method. Use it to find 15% of $46.

T4. A $250 bicycle is sold at a 12% discount. How much money is saved? What is the sale price?

T5. A watch costs $80 before GST. GST is 9%. Mei pays with a $100 note. How much change does she get?

T6. Siti had $200. She spent 40% of it on shoes and 50% of the rest on a bag. How much money did she have left?

T7. Which is bigger: 30% of 70 or 70% of 30? Try another pair like this. What do you notice?

T8. Ravi scored 18 out of 25 in Test A and 27 out of 40 in Test B. In which test did he do better?

CHALLENGE (non-routine: justify, find every way)

X1. A $100 jacket goes up by 20%, then the new price goes down by 20%. What is the final price? Is it a percentage increase or decrease from $100, and how much?

X2. After a 20% discount, a shirt costs $40. Ben says the price before the discount was $48. Is he right? Find the price before the discount.

X3. A tank is 40% full. After 60 ℓ of water is poured in, it is 70% full. How much water can the tank hold?

X4. Mei's score went up by 20% from Test 1 to Test 2. Then it went down by 20% from Test 2 to Test 3. Her Test 3 score was 48. What was her Test 1 score?

X5. A school has 800 pupils. 45% are boys. 20% of the boys and 30% of the girls take the school bus. What percentage of all the pupils take the bus?

X6. A box has 20 marbles. 75% of them are red and the rest are blue. How many red marbles must be taken out so that only 50% of the marbles left are red?

X7. A price tag says $109, GST included. GST is 9%. Ali says the GST is 9% of $109, which is $9.81. Is he right? How much GST is in the price?

X8. An $80 lamp is on sale in two shops. Shop P takes 25% off. Shop Q takes 10% off, and then a further 15% off the new price. In which shop is the lamp cheaper, and by how much?

X9. Mrs Tan spent 40% of her money on groceries and 25% of the remaining money on a gift. She had $54 left. How much money did she have at first?

Lesson 8: Angle Chasers

Angles, Triangles and Quadrilaterals • Week 4

CORE

C1. Two angles on a straight line. One is 127°. What is the other?

C2. Four angles meet at a point. Three are 150°, 90° and 80°. What is the fourth?

C3. Two angles of a triangle are 48° and 67°. What is the third angle?

C4. An isosceles triangle has a top angle of 30°. What is each base angle?

C5. A parallelogram has one angle of 70°. What are the other three angles?

C6. In rhombus ABCD, angle ABC is 124°. What is angle BCD?

THINK HARDER

T1. Two straight lines cross. One of the four angles is 38°. What are the other three? How many different sizes are there?

T2. An isosceles triangle has an angle of 110°. Find the other two angles. Why is there only one answer?

T3. In trapezium ABCD, AB is parallel to DC. Angle DAB is 84° and angle ABC is 97°. Find angle ADC and angle BCD.

T4. ABCD is a square. CDE is an equilateral triangle, and E is outside the square. Find angle ADE and angle DAE.

T5. Rhombus ABCD has angle ABC = 100°. The diagonal AC is drawn. Find angle BAC.

T6. What is the smaller angle between the hands of a clock at 5 o'clock? At 8 o'clock?

T7. How many degrees does the minute hand of a clock turn through in 10 minutes? In 45 minutes?

T8. In a parallelogram ABCD, angle A is 40° bigger than angle B. Find angle A and angle B.

CHALLENGE (non-routine: justify, find every way)

X1. Two identical right-angled triangles with angles 30°, 60° and 90° are joined with their shorter sides (the sides opposite the 30° angles) touching, with the right angles together, to make one big triangle. What are the angles of the big triangle?

X2. Three angles of a four-sided shape are 95°, 110° and 85°. What is the fourth angle? Could the shape be a parallelogram?

X3. In triangle ABC, angle B = 70° and angle C = 50°. The line AD cuts angle A into two equal angles, with D on BC. Find angle ADB and angle ADC.

X4. An isosceles triangle has angles that are all whole numbers of degrees, and one of them is 20°. List every possible triangle.

X5. Triangle ABC has angle A = 20°, angle B = 60° and angle C = 100°. D is a point on AB so that angle ACD = 40°. Find the three angles of triangle ACD and the three angles of triangle BCD.

X6. In parallelogram ABCD, angle ABC = 72°. E is a point on AD with AB = AE. Find angle ABE and angle EBC.

X7. ABC is an isosceles triangle with AB = AC and angle BAC = 40°. D is a point on AC with BD = BC. Find angle ABD.

X8. ABCD is a square. E is a point inside it so that ABE is an equilateral triangle. Find angle DEC.

X9. In an isosceles triangle, one angle is 30° bigger than another angle. Find all the possible sets of three angles.

Lesson 9: Pattern Generalisers

Patterns and Number Sense • Week 5

CORE

C1. 5, 9, 13, 17, … What is the 10th number in this pattern?

C2. Matchstick squares in a row use 4, 7, 10, … sticks. How many sticks are needed for 8 squares?

C3. 2, 6, 10, 14, … Is 50 in this pattern? Explain.

C4. There are 15 bicycles and tricycles in a shed. They have 37 wheels altogether. How many tricycles are there?

C5. The rule for a dot pattern is: number of dots = 3 × figure number + 2. How many dots are in figure 15?

C6. Pens cost $2 each and files cost $3 each. Ben buys 10 items and pays $24. How many files does he buy?

THINK HARDER

T1. Beads repeat in the order red, red, blue, yellow, yellow, yellow. What colour is the 50th bead? How many yellow beads are in the first 100?

T2. 40 square tables in one long row seat 2 × 40 + 2 = 82 people. How many can sit if the tables are put in 8 rows of 5 tables instead?

T3. A farm has chickens and goats. There are 35 heads and 94 legs. How many goats are there?

T4. Work out 1 + 3, 1 + 3 + 5 and 1 + 3 + 5 + 7. What do you notice? Use it to add the first 20 odd numbers, 1 + 3 + … + 39.

T5. Ali saves $3 in week 1. Each week he saves $2 more than the week before. How much does he save in week 15?

T6. 3, 8, 13, 18, … What is the first number in this pattern that is greater than 200?

T7. Two whole numbers next to each other multiply to give 702. What are they?

T8. A test has 10 questions. A correct answer scores 3 marks and 1 mark is taken away for each wrong answer. Siti answered every question and scored 18. How many did she get correct?

CHALLENGE (non-routine: justify, find every way)

X1. A staircase is built from squares: step 1 is 1 square, step 2 adds a column of 2 (3 squares altogether), step 3 adds a column of 3 (6 squares). How many squares are there in the staircase with 10 steps?

X2. Matchstick triangles in a row use 3, 5, 7, … sticks. Which figure uses exactly 99 sticks?

X3. How many squares of all sizes are there on a 4 by 4 grid?

X4. Spiders have 8 legs and beetles have 6 legs. A box holds 20 of them with 136 legs altogether. How many spiders are there?

X5. How many digits are needed to number the pages of a book from page 1 to page 150?

X6. A square grid of matchsticks: a 1 by 1 grid uses 4 sticks and a 2 by 2 grid uses 12 sticks. How many sticks does a 10 by 10 grid use?

X7. A dot pattern makes diamonds. Figure 1 has 1 dot. Figure 2 has rows of 1, 3 and 1 dots (5 dots). Figure 3 has rows of 1, 3, 5, 3 and 1 dots (13 dots). How many dots are in figure 6?

X8. A quiz has 25 questions. A correct answer scores 4 marks, 1 mark is taken away for each wrong answer, and a blank scores 0. Mei left 3 questions blank and scored 68. How many did she get correct?

X9. Ali writes the counting numbers one after another without spaces: 123456789101112131415... What is the 100th digit he writes?

Lesson 10: Challenge Day

Mixed Heuristics • Week 5

CORE

C1. A number is multiplied by 4, then 6 is taken away, and the answer is 50. What was the number?

C2. Ali has 1/2 of his money left after buying a book for $9. How much did he have at first?

C3. Two numbers add up to 40 and the bigger is 8 more than the smaller. What are the two numbers?

C4. Ravi, Siti and Ben each have a different pet: a rabbit, a hamster and a bird. Ravi has no bird and no hamster. Siti has no hamster. Who has the bird?

C5. How many squares of all sizes are in a 2 by 2 grid of small squares?

C6. Pens cost $2 and pads cost $3. Ben buys 10 items and pays $24. How many pads?

THINK HARDER

T1. Siti spent 2/5 of her money, then 1/3 of what was left. She had $24 left at the end. How much did she start with?

T2. How many 3-digit numbers have digits that add up to 5? List them all in order.

T3. The sum of two consecutive numbers (numbers that are next to each other, like 40 and 41) is 145. What are the two numbers?

T4. A snail climbs 3 m up a 10 m wall each day, and slips down 2 m each night. On which day does it reach the top?

T5. Ben has 14 coins, all 50¢ and 20¢ coins, worth $4.90. How many of each?

T6. What is the smallest whole number bigger than 1 that leaves a remainder of 1 when divided by 2, by 3, by 4, by 5 and by 6?

T7. I think of a number, multiply it by 4, take away 10, halve it and add 3. The answer is 22. What is my number?

T8. 1/2 of a number is 5 more than 1/3 of the same number. What is the number?

CHALLENGE (non-routine: justify, find every way)

X1. At the first bus stop, 1/3 of the passengers got off. At the second stop, half of those left got off and 3 got on. There are now 12 passengers. How many were on the bus at first?

X2. How many rectangles of all sizes are there in a 1 by 5 strip of 5 small squares?

X3. Ali, Ben and Mei have 76 stickers altogether. Ali has 3 times as many as Ben. Mei has 6 more than Ben. How many stickers does each have?

X4. How many times does the digit 4 appear when you write all the whole numbers from 1 to 50?

X5. I am a 2-digit number. My digits add up to 11. If you swap my digits, the new number is 27 more than I am. Who am I?

X6. A monkey eats half of the rambutans in a pile and 1 more, every day for 3 days. After the third day there are none left. How many were in the pile at the start?

X7. A water tank is 1/3 full. After 20 litres are added it is 1/2 full. How many litres does the full tank hold?

X8. 25 coins of 50¢, 10¢ and 20¢ are worth $5.30 altogether. There are twice as many 10¢ coins as 50¢ coins. How many coins of each kind are there?

X9. Ali, Ben, Mei and Siti sit in a row of 4 chairs. Mei is not on either end. Ben does not sit next to Mei. Siti sits somewhere to the left of Ali. Ben is not on the left end. In what order do they sit, from left to right?

Answer key (tutor only)

Lesson 1: Million Puzzles

C1. Four million, thirty thousand, two hundred and five.

C2. 600 000

C3. 5 483 000

C4. 2750

C5. 6 (brackets first: 6 × 7 = 42, then 48 − 42)

C6. About 350 000 (7000 × 50). The exact answer is 331 776.

T1. 7 numbers: 1 000 001, 1 000 010, 1 000 100, 1 001 000, 1 010 000, 1 100 000 and 2 000 000.

T2. (2 + 3) × 4 + 5 = 25 and 2 + 3 × (4 + 5) = 29. (29 is the biggest you can make with one pair.)

T3. Greatest 640 499, smallest 639 500.

T4. 6 419 754 (7 654 321 − 1 234 567).

T5. 120 000, because 4000 × 30 = 120 000. (Exactly 119 132.)

T6. 4000 weeks. With 52 weeks in a year, that is about 77 years.

T7. (4 + 4 + 4) ÷ 4 = 3; (4 × 4 + 4) ÷ 4 = 5; 4 + 4 − 4 ÷ 4 = 7. Other ways are possible.

T8. Smallest 1 023 456; greatest 9 876 543.

X1. 9 000 000. They go from 1 000 000 to 9 999 999, and 9 999 999 − 1 000 000 + 1 = 9 000 000.

X2. 64 × 15 625. (1 000 000 = 10 × 10 × 10 × 10 × 10 × 10; put all the 2s together to make 64 and all the 5s together to make 15 625.)

X3. 6 or 7. The smallest answer is 1000 × 100 = 100 000 (6 digits); the greatest is 9999 × 999 = 9 989 001 (7 digits).

X4. 1000. The first and last digits are 3; the 2nd, 3rd and 4th digits can be anything (10 × 10 × 10), and they fix the rest.

X5. (1 × 2 + 3) × 4 × 5 = 100 (or 1 × (2 + 3) × 4 × 5 = 100).

X6. About 10 days. 70 × 60 × 24 = 100 800 beats a day, and 1 000 000 ÷ 100 800 is just under 10.

X7. 231: (1 + 2) × (3 + 4) × (5 + 6) = 3 × 7 × 11. With no brackets it is only 33.

X8. 6 953 487. Seven different digits adding to 42 must be 3, 4, 5, 6, 7, 8, 9 (the biggest seven digits add to exactly 42), so there is no 0. The number is between 6 950 000 and 7 049 999, so it starts 6 9 with 5, 7 or 8 next; the smallest is 6 95 then 3, 4, 8, 7 (it must end in an odd digit).

X9. 2893. 9 × 1 + 90 × 2 + 900 × 3 + 4 = 9 + 180 + 2700 + 4. Go further: 5 888 896 (9 × 1 + 90 × 2 + 900 × 3 + 9000 × 4 + 90 000 × 5 + 900 000 × 6 + 7 = 9 + 180 + 2700 + 36 000 + 450 000 + 5 400 000 + 7).

Lesson 2: Fraction Feats

C1. 7/9

C2. 17/12 = 1 5/12

C3. 7/12

C4. 9

C5. 3/5

C6. 2/5 (3/10 + 1/10 = 4/10)

T1. 1/2 + 1/6 (the only way).

T2. 1/2. 2/3 was left after Ben; Ali ate 1/4 of it, so 3/4 of 2/3 = 1/2 is left.

T3. 1/20 and 1/90 (the bottom numbers multiply: 4 × 5 and 9 × 10).

T4. 4/6 = 2/3 of a litre.

T5. 6/7. 5/6 is 1/6 short of 1 and 6/7 is only 1/7 short of 1, and 1/7 is smaller than 1/6.

T6. 1/4 m (3/4 ÷ 3).

T7. 9/20 (2/5 = 8/20 and 1/2 = 10/20).

T8. 5/6 of 30. 7/8 of 24 = 21 and 5/6 of 30 = 25.

X1. Only one way: 1/2 + 1/3 + 1/6. (The biggest part must be 1/2, and then 1/2 = 1/3 + 1/6 is the only split into two different unit fractions.)

X2. 1/100. Each top number cancels the bottom number before it, leaving 1 on top and 100 at the bottom.

X3. 120. What is left is 1/2 × 2/3 × 3/4 = 1/4 of the cookies, and 1/4 is 30.

X4. They are the same: both are 1/2 of the cake (2/3 × 3/4 = 3/4 × 2/3 = 6/12).

X5. 8/35, from 3/7 − 1/5 (15/35 − 7/35).

X6. $25 each day. 1/3 of the remaining 3/4 is 1/4, so she spent 1/4 each day and 1/2 is left. 1/2 = $50, so she had $100.

X7. 5/14. Each fraction is a difference: (1/2 − 1/3) + (1/3 − 1/4) + (1/4 − 1/5) + (1/5 − 1/6) + (1/6 − 1/7). The middle parts cancel, leaving 1/2 − 1/7 = 5/14.

X8. Each child gets 5/12. Two ways: 1/3 + 1/12 (cut 4 pizzas into thirds and 1 pizza into twelfths), or 1/4 + 1/6 (cut 3 pizzas into quarters and 2 pizzas into sixths).

X9. 54 litres. Each time 2/3 is left: 54 → 36 → 24 → 16.

Lesson 3: Before and After

C1. 300 (left = 2/3 of 3/5 = 2/5 of all the buns; 2/5 = 120, so 1/5 = 60). She sold 60 in the afternoon.

C2. 18 years (the difference, 28, is 1 unit; Ben will be 28 and his father 56)

C3. $30 (book $15, remainder $45, T-shirt $15)

C4. 4 (the total is 60, so 30 each)

C5. 10 (15 boys and 25 girls)

C6. $54 (after: the difference $24 is 2 units, so 1 unit = $12; Siti $12 + $42 = $54; Mei $36 + $18 = $54)

T1. $120 (20 units: Monday 5 units, Tuesday 6 units; 11 units = $66, so 1 unit = $6)

T2. 36 (the red pens did not change: red 1 unit, blue 2 units at first, 1 unit after; 1 unit = 12)

T3. 1 year ago (the difference, 32, is 4 units; 1 unit = 8: son 8, Mrs Tan 40)

T4. $80 (Ben 2 units, Siti 1 unit + $20; 3 units = $120, 1 unit = $40)

T5. 50 (the difference, 40, is 2 units; Mei 3 units = 60 after, so 50 at first)

T6. 72 litres (1/4 of the tank = 18 litres)

T7. 45 (Siti 3 units; after giving 1 unit she has 2 units, and Mei has 15 + 1 unit = 2 units, so 1 unit = 15)

T8. $15 (the difference, $30, is 1 unit; Ben $30, Ravi $60)

X1. $30 (Ali $60, Ben $30, Mei $90)

X2. $48 (work backwards: 10 + 8 = 18; 18 × 2 = 36; 36 + 12 = 48)

X3. 10 (in 5 years the total is 60 = 4 units, the son is 15)

X4. 160 (Ali 3 units, Ben 5 units; same means 4 units each, so Ben gave 1 unit = 20; 8 units = 160)

X5. 60 (the total did not change; 1/3 − 1/6 = 1/6 of the fruits is 10)

X6. $150 (15 units: dress 6 units, remainder 9 units, shoes 3 units; 3 units = $30, 1 unit = $10)

X7. $60 (Siti spent 1 unit, Mei spent 3 units; the $120 difference is 2 units, so 1 unit = $60; both have $20 left)

X8. 40 (work backwards: (4 + 4) × 2 = 16, (16 + 4) × 2 = 40)

X9. 40 (Ali 3 units, keeps 2 units; Ben keeps 2 units, so Ben had 4 units; 7 units = 140, 1 unit = 20)

Lesson 4: Triangle Areas

C1. 30 cm²

C2. 60 cm² (1/2 × 8 × 15; 17 cm is the slanting side)

C3. 9 cm (1/2 × 9 × 8 = 36)

C4. 27 cm²

C5. 32 cm²

C6. 126 cm² (63 + 63)

T1. 9 cm

T2. 96 cm² (1/2 × 12 × 16)

T3. 50 cm²

T4. 42 cm² (84 − 42)

T5. 10 cm (area 20 cm²)

T6. 48 cm² and 48 cm²

T7. 48 cm² (36 + 12)

T8. 15 cm²

X1. 30 cm² (half the base, same height)

X2. 36 cm² (96 − 24 − 12 − 24)

X3. 8 cm (1/2 × 9 × 8 = 36)

X4. 16 cm² (add the shape ABCF to both: square 64 cm² and triangle ABE 48 cm², so the difference is 64 – 48)

X5. 6 pairs: 1 and 12, 2 and 6, 3 and 4, 4 and 3, 6 and 2, 12 and 1 (base × height must be 12)

X6. 24 cm (Mei's area is 60 cm²)

X7. 12 cm (area 1/2 × 15 × 20 = 150 cm², and 1/2 × 25 × 12 = 150)

X8. 50 cm² both times: it is always half of the big square (box it in a rectangle and take away three right-angled triangles: 160 – 50 – 12 – 48 = 50, and 140 – 50 – 12 – 28 = 50)

X9. 18 cm² (the height is 9 cm because 1/2 × 12 × 9 = 54; ABD has base 4 cm and the same height)

Lesson 5: Volume Vault

C1. 180 cm³

C2. 216 (6 × 6 × 6)

C3. 12 ℓ (40 × 20 × 15 = 12 000 cm³)

C4. 5 cm (240 ÷ 48)

C5. 10 cm (2000 ÷ 500 = 4 cm more)

C6. 600 cm³ (20 × 15 × 2)

T1. 24 (4 along, 3 across, 2 layers; not 270 ÷ 8, because the cubes cannot be cut)

T2. 45 ℓ (3/5 – 1/3 = 4/15 of the tank is 12 ℓ, so 1/15 is 3 ℓ)

T3. 7 cm (base area 245 ÷ 5 = 49 cm², and 7 × 7 = 49)

T4. 6 (the inside block is 1 × 2 × 3)

T5. 12 buckets (3/4 of 120 ℓ = 90 ℓ; 90 ÷ 8 = 11.25, so 11 buckets are not enough)

T6. 30 cm (36 000 ÷ 1200)

T7. 5 cm (8000 ÷ 1600)

T8. 125 (5 × 5 × 5)

X1. 216 (each face has 96 ÷ 6 = 16 = 4 × 4 one-face cubes, so the cube is 6 cubes along each edge: 6 × 6 × 6)

X2. 7 (1 × 1 × 64, 1 × 2 × 32, 1 × 4 × 16, 1 × 8 × 8, 2 × 2 × 16, 2 × 4 × 8, 4 × 4 × 4)

X3. 6600 cm³ = 6.6 ℓ (40 × 30 × 6 – 10 × 10 × 6 = 7200 – 600; the cube is taller than the water, so only 6 cm of it is under water)

X4. 22.5 cm (each brick raises it 3 ÷ 4 = 0.75 cm; 10 bricks raise it 7.5 cm)

X5. 98 (125 – 3 × 3 × 3)

X6. 72 (the smallest cube edge is 12 cm, the smallest common multiple of 2, 3 and 4: 6 × 4 × 3 = 72 bricks)

X7. 60 cm³ (edges 3 cm, 4 cm and 5 cm: 3 × 4 = 12, 3 × 5 = 15, 4 × 5 = 20)

X8. 24 with 2 faces (4 × 1 + 4 × 2 + 4 × 3 along the edges); 22 with 1 face (60 – 8 corners – 24 – 6 inside)

X9. 12 cm (the water now covers 600 – 100 = 500 cm² of floor: 6000 ÷ 500 = 12 cm; the block is 15 cm tall, so it sticks out. Not 12.5 cm!)

Lesson 6: Rate Racers

C1. $5.10 ($1.70 a kg)

C2. 15 minutes

C3. 6 rolls for $4.98 (83¢ a roll, against 85¢)

C4. 15 minutes

C5. 8 minutes (192 ÷ 24)

C6. $9.75

T1. 5 ℓ a minute (the tank holds 180 ℓ; 180 ÷ 9 = 20 ℓ a minute together; 20 – 15 = 5)

T2. $9.60 (one box of 10 and three boxes of 6 make 28 eggs; three boxes of 10 cost $9.90)

T3. $9.50 (4 h 35 min: first hour, then 3 h 35 min is 8 half hours or parts)

T4. 100 minutes (Plan B costs 15¢ more each minute; $15 ÷ 15¢ = 100; both cost $25)

T5. 900 (1 machine makes 30 masks a minute: 30 × 5 × 6)

T6. $240 ($180 + $60)

T7. 30 minutes (300 ÷ 10)

T8. 28 ℓ

X1. 2 minutes (pretend the job is 6 pages: P prints 1 page a minute, Q 2 pages, together 3 a minute)

X2. 30 pens (6 pens cost $4 and sell for $6: $2 profit for every 6 pens)

X3. 10.15 a.m. (400 cups by 10.00; 1200 ÷ 80 = 15 minutes more)

X4. $13.64 (1 deal from C and 2 packs from B make 20 cans). Check the bigger totals: 22 cans cost at least $14.92 (C + A + B) and 24 cans cost at least $15.60 (2 deals from C), so they cannot be cheaper.

X5. 15 minutes (135 000 cm³ = 135 ℓ; 135 ÷ 9)

X6. 90 pages (Ravi takes 15 hours; Siti reads 270 pages in that time)

X7. 3 cats (each cat eats 1 fish every 3 minutes, so the 3 cats eat 1 fish a minute: 100 fish in 100 minutes)

X8. After 6 hours, both 8 cm tall (the gap of 6 cm closes by 1 cm each hour)

X9. 30 minutes (pretend the bath holds 60 ℓ: 5 ℓ a minute in, 3 ℓ a minute out, 2 ℓ a minute stay: 60 ÷ 2)

Lesson 7: Percentage Puzzles

C1. $36 and $54 ($36 + $18)

C2. $36 (20% of $45 = $9)

C3. $32.70 (9% of $30 = $2.70)

C4. 60% (24 out of 40 = 60 out of 100)

C5. $45

C6. 35% (21 out of 60 = 35 out of 100)

T1. (a) 25% ($1 out of $4) (b) 20% ($1 out of $5) (c) Each change is compared with its own starting amount: $1 out of $4 is a bigger part than $1 out of $5. (Percentage change is a P6 skill: a preview.)

T2. (a) Shop X $45 (she pays for 3 boxes); Shop Y $42 (30% of $60 = $18 off). Shop Y is the better buy, by $3. (b) 25% off (1 box free out of 4, $15 off $60)

T3. She found 10% ($8) and 5% (half of 10%, $4) and added them. 15% of $46 = $4.60 + $2.30 = $6.90.

T4. Saved $30 (10% = $25, 2% = $5); sale price $220.

T5. $12.80 (GST $7.20, so the watch costs $87.20)

T6. $60 (shoes $80, $120 left, bag $60, $60 left)

T7. They are the same: both are 21. In general, a% of b is the same as b% of a (for example 4% of 25 = 25% of 4 = 1).

T8. Test A: 72% (18 out of 25) is more than 67.5% (27 out of 40).

X1. $96 ($120, then 20% of $120 = $24 off). A 4% decrease.

X2. No. $40 is 80% of the old price, so 10% is $5 and 100% is $50. (Ben added 20% of $40, but the 20% was of the old price.)

X3. 200 ℓ (the 60 ℓ is 30% of the tank, so 10% is 20 ℓ)

X4. 50. Work backwards: 48 is 80% of Test 2, so Test 2 = 60; 60 is 120% of Test 1, so Test 1 = 50.

X5. 25.5% (boys 360, 72 take the bus; girls 440, 132 take the bus; 204 out of 800)

X6. 10. The 5 blue marbles stay. If they are 50% of the box, there are 10 marbles left, so 20 − 10 = 10 red marbles are taken out.

X7. No. $109 is 109% of the price before GST, so 1% is $1 and the GST is $9. (9% of $109 is wrong because GST is 9% of the price BEFORE GST.)

X8. Shop P, by $1.20. P: $80 − $20 = $60. Q: $80 − $8 = $72, then 15% of $72 = $10.80 off, $61.20.

X9. $120. Work backwards: $54 is 75% of the remainder, so the remainder is $72. $72 is 60% of her money, so 10% is $12 and 100% is $120.

Lesson 8: Angle Chasers

C1. 53° (180° − 127°)

C2. 40° (360° − 320°)

C3. 65° (180° − 115°)

C4. 75° ((180° − 30°) ÷ 2)

C5. 110°, 70°, 110° (opposite angles are equal; adjacent angles add up to 180°)

C6. 56° (adjacent angles add up to 180°)

T1. 142°, 38°, 142°. There are two different sizes: 38° and 142°.

T2. 35° and 35°. 110° cannot be a base angle (two of them would be 220°), so it is the top angle and 180° − 110° = 70° is shared by two equal base angles.

T3. ADC = 96° and BCD = 83° (angles between the parallel sides add up to 180°). Check: 84° + 97° + 96° + 83° = 360°.

T4. ADE = 150° (90° + 60°). Triangle ADE is isosceles (AD = DE), so DAE = (180° − 150°) ÷ 2 = 15°.

T5. 40° (AB = BC, so triangle ABC is isosceles: (180° − 100°) ÷ 2)

T6. 150° at 5 o'clock (5 × 30°). At 8 o'clock the big angle is 240°, so the smaller angle is 120°.

T7. 60° (a full turn of 360° takes 60 minutes, so 6° a minute) and 270°.

T8. A = 110°, B = 70°. Adjacent angles add up to 180°; 180° − 40° = 140°, and 140° ÷ 2 = 70°, so A is 70° + 40° = 110°.

X1. 30°, 30°, 120°. The two right angles make a straight line, and the two 60° angles together make 120°.

X2. 70° (360° − 290°). It cannot be a parallelogram: the angles 95°, 110°, 85° and 70° have no equal opposite pairs.

X3. Angle A = 60°, so each part is 30°. ADB = 180° − 70° − 30° = 80°, and ADC = 180° − 80° = 100° (check: 30° + 50° + 100° = 180°).

X4. 20°, 20°, 140° and 20°, 80°, 80°.

X5. Triangle ACD: 20°, 40°, 120°. Triangle BCD: angle BCD = 60°, B = 60° and D = 60°, so it is equilateral.

X6. Angle BAD = 108° (adjacent angles add up to 180°). Triangle ABE is isosceles with top angle 108°, so ABE = 36°. Then EBC = 72° − 36° = 36°.

X7. 30°. Angles B and C are (180° − 40°) ÷ 2 = 70°. In triangle BCD, BD = BC, so angle BDC = angle C = 70°, which makes angle CBD = 40°. Then ABD = 70° − 40° = 30°.

X8. 150°. AE = AB = AD, so triangle ADE is isosceles with angle DAE = 90° − 60° = 30°, which makes angle AED = 75°. In the same way angle BEC = 75°. Angle AEB = 60°. Round E: 360° − 75° − 75° − 60° = 150°.

X9. Two sets: 50°, 50°, 80° and 40°, 70°, 70°. (If the top angle is 30° bigger than a base angle: 3 equal parts and 30° make 180°, so base 50°. If a base angle is 30° bigger than the top: top 40°, base 70°.)

Lesson 9: Pattern Generalisers

C1. 41 (5 + 9 × 4: there are 9 jumps of 4 from the 1st number to the 10th).

C2. 25 (3 × 8 + 1).

C3. Yes. 50 − 2 = 48 and 48 ÷ 4 = 12 jumps, so 50 is the 13th number.

C4. 7 tricycles (suppose all 15 are bicycles: 30 wheels; 7 wheels short; each tricycle adds 1 more wheel). 8 bicycles. Check: 16 + 21 = 37.

C5. 47

C6. 4 files and 6 pens (all pens would be $20; each file instead of a pen adds $1). Check: $12 + $12 = $24.

T1. Red (50 = 8 × 6 + 2, so it is the 2nd bead of a group). 49 yellow beads: 16 full groups of 6 give 96 beads and 48 yellow, and beads 97 to 100 are red, red, blue, yellow.

T2. 96 (each row of 5 seats 12, and 8 × 12 = 96). One long row seats only 82. Splitting the tables gives more end seats.

T3. 12 goats and 23 chickens (all chickens would have 70 legs; 24 legs short; each goat adds 2 legs). Check: 48 + 46 = 94.

T4. 4, 9, 16: square numbers. The first 20 odd numbers add up to 20 × 20 = 400. (Each new odd number adds an L-shaped layer to a square of dots.)

T5. $31 ($3 + 14 × $2).

T6. 203 (the 41st number: 3 + 40 × 5).

T7. 26 and 27. 25 × 26 = 650 is too small and 27 × 28 = 756 is too big. Check: 26 × 27 = 702.

T8. 7 (all correct would be 30; each wrong answer costs 3 + 1 = 4 marks; 12 ÷ 4 = 3 wrong). Check: 21 − 3 = 18.

X1. 55 (1 + 2 + 3 + ... + 10). Two copies of the staircase make a 10 by 11 rectangle: 110 ÷ 2 = 55.

X2. Figure 49 (sticks = 2 × figure number + 1; 99 − 1 = 98, 98 ÷ 2 = 49).

X3. 30 (16 small, 9 of size 2 by 2, 4 of size 3 by 3 and 1 of size 4 by 4).

X4. 8 spiders and 12 beetles (all beetles: 120 legs; 16 short; each spider adds 2). Check: 64 + 72 = 136.

X5. 342 (9 one-digit pages, 90 two-digit pages = 180 digits, 51 three-digit pages = 153 digits).

X6. 220. There are 11 rows of 10 sticks going across and 11 columns of 10 sticks going down: 110 + 110.

X7. 61. The counts go 1, 5, 13, 25, 41, 61 (add 4, 8, 12, 16, 20). Or: each diamond is two square numbers of dots, 6 × 6 + 5 × 5 = 36 + 25 = 61.

X8. 18. She answered 22 questions. All 22 correct would be 88 marks; each wrong answer costs 4 + 1 = 5 marks; 88 − 68 = 20, 20 ÷ 5 = 4 wrong. Check: 72 − 4 = 68.

X9. 5. The numbers 1 to 9 use 9 digits. That leaves 91 digits: 45 two-digit numbers (10 to 54) use 90 of them, so the 100th digit is the first digit of 55.

Lesson 10: Challenge Day

C1. 14 (work backwards: 50 + 6 = 56, 56 ÷ 4 = 14).

C2. $18 ($9 is the other half).

C3. 24 and 16 (take away the 8: 32, share: 16 and 16, then add 8 back to one: 24 and 16).

C4. Siti. Ravi has the rabbit (no bird, no hamster). Siti is not the hamster, so she has the bird, and Ben has the hamster.

C5. 5 (4 small and 1 big).

C6. 4 pads and 6 pens (all pens would be $20; each pad instead of a pen adds $1).

T1. $60. Work backwards: $24 is 2/3 of what was left, so that was $36; $36 is 3/5 of her money, so 1/5 is $12 and she started with $60. Check: 60 − 24 = 36, 36 − 12 = 24.

T2. 15 numbers: 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500.

T3. 72 and 73 (145 is 1 more than twice 72).

T4. Day 8. After 7 days and nights it is at 7 m. On day 8 it climbs 3 m to reach 10 m before it can slip.

T5. 7 fifty-cent coins and 7 twenty-cent coins (all 20¢ would be $2.80; $2.10 short; each swap adds 30¢; 210 ÷ 30 = 7).

T6. 61 (it is 1 more than a multiple of 2, 3, 4, 5 and 6; the smallest such multiple is 60).

T7. 12. Work backwards: 22 − 3 = 19, 19 × 2 = 38, 38 + 10 = 48, 48 ÷ 4 = 12.

T8. 30. The difference between a half and a third is 1/6, and 1/6 of the number is 5. Check: 15 and 10.

X1. 27. Work backwards: 12 − 3 = 9; 9 is half of what was left, so 18 were left after stop 1; 18 is 2/3 of the passengers, so 1/3 is 9 and there were 27 at first.

X2. 15 (5 of length 1, 4 of length 2, 3 of length 3, 2 of length 4 and 1 of length 5: 5 + 4 + 3 + 2 + 1).

X3. Ben 14, Ali 42, Mei 20 (Ben 1 unit, Ali 3 units, Mei 1 unit and 6: 5 units and 6 = 76; 5 units = 70; 1 unit = 14).

X4. 15 (in the ones place: 4, 14, 24, 34 and 44 is 5 times; in the tens place: 40 to 49 is 10 times; 5 + 10 = 15).

X5. 47 (74 − 47 = 27; the digits are 4 and 7).

X6. 14. Work backwards: after day 3 there are 0; before day 3 there were 2 (half of 2 is 1, and 1 more is 2); before day 2: (2 + 1) × 2 = 6; before day 1: (6 + 1) × 2 = 14. Check: 14 → 6 → 2 → 0.

X7. 120 litres. The 20 litres are 1/2 − 1/3 = 1/6 of the tank, so the whole tank is 6 × 20 = 120 litres.

X8. 3 fifty-cent coins, 6 ten-cent coins and 16 twenty-cent coins. Suppose all 25 coins were 20¢: that is $5.00. Every 50¢ coin comes with two 10¢ coins; this group of 3 coins is worth 70¢, and replacing three 20¢ coins (60¢) with it adds 10¢. The extra is 30¢, so there are 3 groups. Check: 150¢ + 60¢ + 320¢ = 530¢ and 3 + 6 + 16 = 25.

X9. Siti, Mei, Ali, Ben. Mei is in chair 2 or 3. If Mei is in chair 3, the only chair not next to her is chair 1, but Ben is not on the left end, so that fails. So Mei is in chair 2, the only chair not next to her is chair 4, so Ben is in chair 4. Siti and Ali are in chairs 1 and 3 and Siti is on the left: Siti 1, Ali 3.